

Section A

Answer all questions in the spaces provided.

Use the information in the separate Resource Folder to answer the following questions.

1. (a) (i) Use the information in **Table 1** to describe the trend in emissions of HFC between 1990 and 2014. [2]

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- (ii) Use the information in **Table 1** to plot a graph of net CO₂ emissions from the UK between 1990 and 2014. [4]

Net CO₂
Emissions
(MtCO₂eq)



- (iii) Describe the trend shown in the graph. [2]

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- (iv) Describe how your graph (opposite) compares with the information in **Graph 1**. [2]

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- (v) Use your knowledge of the carbon cycle to account for the seasonal fluctuations in CO₂ concentration shown in **Graph 1**. Give reasons for your answer. [3]

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- (b) Use the information in **Tables 2** and **3** and the equation:

$$\text{greenhouse contribution} = (\text{concentration in the atmosphere}) \times (\text{multiplier})$$

to complete the table below.

[3]

gas	greenhouse contribution					
	multiplier	pre-industrial baseline	natural additions	man-made additions	total relative contribution	% of total contribution
carbon dioxide	1	288 000		11 880	368 400	72.37
methane	21	17 808	12 117		36 645	

- (c) Use the information in **Tables 2** and **4** to describe the effect on the data in the last column of **Table 2** when water vapour is included. [2]

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- (d) State **one** conclusion that can be made from the information shown in **Chart 1** on page 4 of the resource folder. [1]

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Examiner
only

- (e) Explain whether **Graph 2** and **Graph 3** support the view that global warming is due to solar activity and not CO₂ concentration. [6 QER]